

4- Bayesian approximated inference: Laplace approximation

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Overview

- 1- The method
- 2- Examples of application

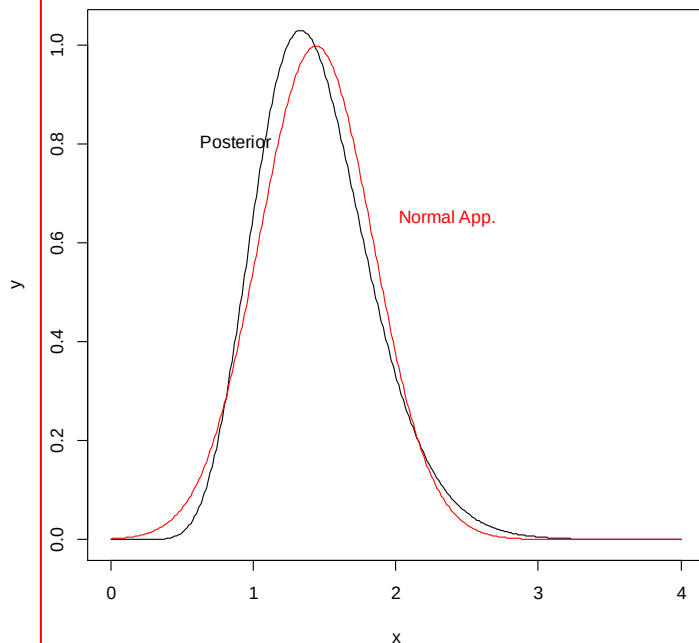
An introductory example

Consider a Poisson model,

$$P(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!} \quad x=0,1,2,\dots \text{ and } \lambda>0. \text{ We have } n=8 \text{ observations with mean equal to } 1.5$$

Consider a prior $\pi(\lambda) \approx \text{Gamma}(1,1)$

It is easy to verify that the posterior is $f(\lambda) \approx \text{Gamma}(13,9)$ (check this!)



Can the posterior be approximated by a normal distribution?

A simple way is to use a normal distribution with the same mean and variance that the posterior.
In this case, $\mu = 13/9 = 1.44$ and $\sigma^2 = 0.16$

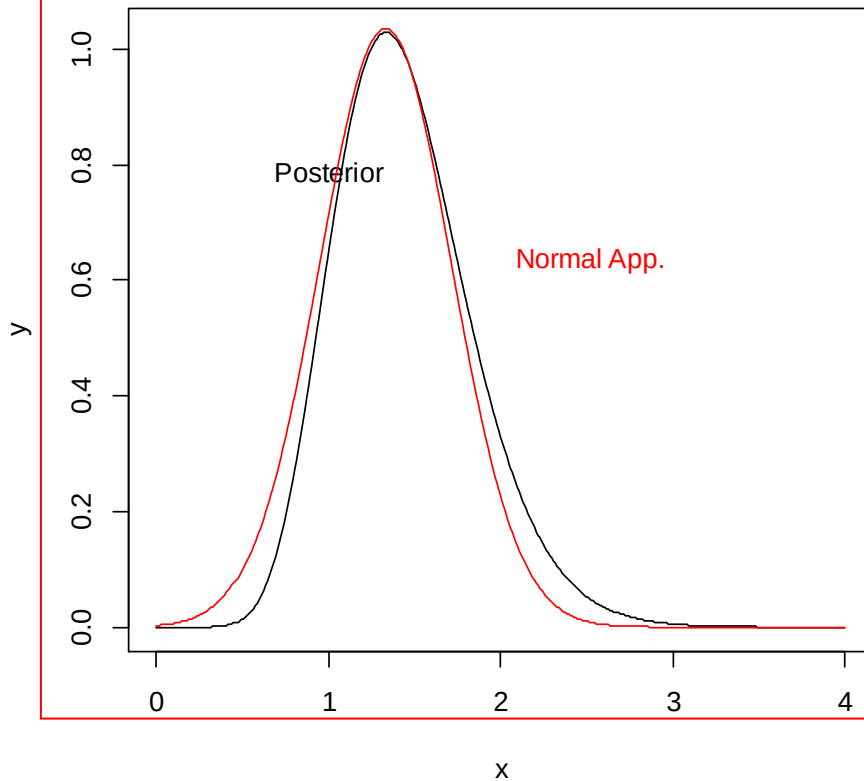
Another approximation can be obtained from a kind of modified log-likelihood:

$$l^*(\lambda) = \log(f(\lambda)) \approx \log(L(\lambda)) + \log(\pi(\lambda))$$

This is, $l^*(\lambda) \sim 12 \log(\lambda) - 9\lambda$

The MLE is, $\frac{dl^*}{d\lambda} = \frac{12}{\lambda} - 9 = 0 \rightarrow \lambda^* = 12/9 = 1.33$

The asymptotic estimated variance, $V(\lambda^*) = \lambda^{*2}/12 = 0.1481$



Now the normal approximation with $\mu = 1.33$ and $\sigma^2 = 0.1481$ is better!!

1- Laplace approximation: the method

(based on [Bernstein–von Mises theorem](#))

Let θ , in $\mathbb{R}^k$, be the vector of parameters and $f(\theta|X) \propto L(X|\theta)\pi(\theta)$

Let θ^* be the maximum (the mode) of $f(\theta|X)$. Consider $g(\theta) = \log(f(\theta|X))$. The second order Taylor series expansion about θ^* is,

$$g(\theta) \sim g(\theta^*) + \sum_{i=1}^k \frac{\partial g}{\partial \theta_i}(\theta^*)(\theta_i - \theta_i^*) - \frac{1}{2}(\theta - \theta^*)^t \cdot V(\theta^*) \cdot (\theta - \theta^*)$$

where $V(\theta)$ is a matrix with elements equal to $-\frac{\partial^2 g}{\partial \theta_i \partial \theta_j}$

Taking into account that $\frac{\partial g}{\partial \theta_i}(\theta^*) = 0$ (because this is a maximum), we obtain,

$$f(\theta|X) \propto L(X|\theta^*)\pi(\theta^*)\exp\left(-\frac{1}{2}(\theta - \theta^*)^t V(\theta^*)(\theta - \theta^*)\right)$$

Note that the right part is proportional to a k -dimensional multivariate normal distribution with mean θ^* and variance matrix $V(\theta^*)^{-1}$.

Note that, in some sense, THE PROBLEM IS EQUIVALENT TO a *penalized likelihood* where the log-likelihood $\log(L(\theta))$ is replaced by $\log(L(\theta)) + \log(\pi(\theta))$.

Consequently, the procedure to make inference with the Laplace approximation is the following:

- 1- Compute the log-likelihood $\log(L(X|\theta))$
- 2- Include the information of the prior constructing a new log-likelihood function (this is a penalized likelihood): $l^*(\theta) = \log(L(X|\theta)) + \log(\pi(\theta))$
- 3- Make classical inference with $l^*(\theta)$, calculating the MLE θ^* and the asymptotic variance matrix Σ^* (by means of the hessian).
- 4- The posterior distribution is approximately $f(\theta|X) \sim N(\theta^*, \Sigma^{*-1})$

Example (from the book Estadística II: Inferencia, Ruiz-Maya, p. 534)

We have two independent samples coming from $N(\mu_1, \sigma)$ and $N(\mu_2, \sigma)$:

$x_1 = c(181, 145, 150, 123, 156, 160, 115, 150, 134)$

$x_2 = c(121, 144, 123, 116, 156, 142, 101)$

We want to make inference on the difference of the population means $d = \mu_1 - \mu_2$

The likelihood is $L(X | \mu_1, \sigma, d) = L(X_1 | \mu_1, \sigma) L(X_2 | \mu_1 - d, \sigma)$

The MLE are $\hat{\mu}_1 = \bar{x}_1$, $\hat{d} = \bar{x}_1 - \bar{x}_2$ and $\hat{\sigma} = \sqrt{\frac{s_1^2(n_1-1) + s_2^2(n_2-1)}{n_1 + n_2}}$



The values are, $\hat{\mu}_1 = 146$, $\hat{d} = 17$, $s_1 = 19.96$, $s_2 = 19.04$ and $\hat{\sigma} = 18.919$

We consider a prior assuming that the three parameters are independent,
 $\pi(\mu, d, \sigma) = h(\mu)z(d)w(\sigma)$ (note that for simplicity we call μ_1 as μ).

We are convinced that μ has to be positive, then a Gamma prior with mean equal to 146 and standard deviation 20 can be a good option (informative prior).

It leads to a Gamma(53.29,0.365)



Because we have no preference between the two means, the prior for d could be $N(0,40)$. On the other hand, it seems that there are not many changes for the standard deviation of each sample. Then, we assume σ to be uniform in $[18,21]$.

The r code for doing the Laplace approximation is the following,

```
x1=c(181, 145, 150, 123, 156, 160, 115, 150, 134)
x2=c(121,144,123,116,156,142,101)
```

```
# Definition of the (-) penalized log-likelihood
```

```
pllike=function(theta){-sum(log(dnorm(x1,theta[1], theta[3])))-
sum(log(dnorm(x2,theta[1]-theta[2], theta[3])))-log(dgamma(theta[1],
53.29,0.365))-log(dnorm(theta[2],0,40))}
```

```
pMLE=nlm(pllike,c(146,17,18.9),hessian=T) #Starting points are the MLE
```

The estimated variance matrix is just,
`solve(pMLE$hessian)`

	[,1]	[,2]	[,3]
[1,]	33.4734453	32.528416	-0.5658985
[2,]	32.5284160	78.176235	-1.0805985
[3,]	-0.5658985	-1.080598	10.5063110

Then, the posterior marginal distribution of d can be approximated by a normal distribution $N(15.948, 8.842)$. Consequently, a 95% credibility interval is just 15.948 ± 17.330 , that is $[-1.382, 33.278]$. On the other hand, $P(d>0) \sim 0.964$ (`1-pnorm(0,15.948, 8.842)`). Have we used here the prior for σ ? 

Note that $P(d>0)/(1 - P(d>0)) = 26.8$, showing a strong evidence in favor of $d>0$.

```
> pMLE
$minimum
[1] 77.82884

$estimate
[1] 145.42564 15.94786 18.31762

$gradient
[1] -1.596729e-07 9.712797e-08 -1.171462e-07

$hessian
      [,1]      [,2]      [,3]
[1,] 0.0501570404 -0.020862202 0.0005558686
[2,] -0.0208622019 0.021487198 0.0010863133
[3,] 0.0005558686 0.001086313 0.0953225576

$code
[1] 1

$iterations
[1] 8

> |
```

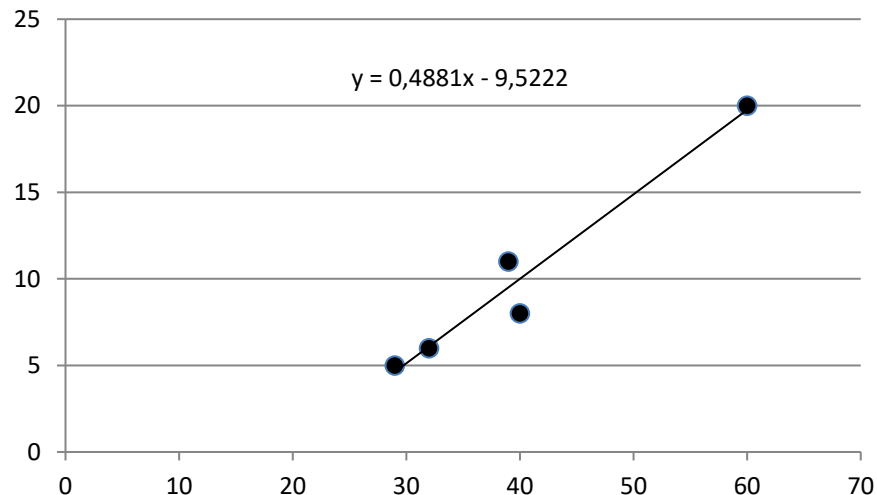
For thinking: If the prior is absolutely noninformative $\pi(\theta) \sim 1$, then the Laplace approximation leads to the classical MLE with the classical asymptotic distribution... Where is the Bayesian inference here? 

Exercise 1

The first row is chest circumference (in inches) of five subjects. Let us call this X . The second row is the respective total volumes of air that can be breathed in and out in one minute (in liters) for the same five subjects. Let us call this Y .

X : 39, 29, 60, 40, 32

Y : 11, 5, 20, 8, 6



We want to estimate and make inference on the slope and intercept of the regression line $Y_i = \alpha + \beta X_i + e_i$, where e_i are iid $N(0, \sigma)$.

The prior information that we have comes from a previous large study where the fitted line was $Y = -10 + 0.5 X$ with standard errors of 1.3 and 0.03.

The likelihood function is, $L(Y|\alpha, \beta, \sigma) = \prod_{i=1}^5 \text{dnorm}(y_i, \alpha + \beta x_i, \sigma)$

The prior for α will be a normal distribution $N(-10, 1.3)$ and the prior for β will be $N(0.5, 0.03)$.

```
x=c(39,29,60,40,32)
```

```
y=c(11,5,20,8,6)
```

```
# Definition of the (-) penalized log-likelihood
```

```
pllike=function(theta){-sum(log(dnorm(y,theta[1]+theta[2]*x, theta[3])))-  
log(dnorm(theta[1],-10,1.3))-log(dnorm(theta[2],0.5,0.03))}
```

```
pMLE=nlm(pllike,c(-9.52,0.49,1.46),hessian=T) #Start. points are the MLE
```

```
pMLE
```

```
diag(solve(pMLE$hessian))^0.5
```

Consequently, $\alpha \sim N(-9.921, 0.86)$
 $\beta \sim N(0.498, 0.02)$

The classical procedure using `lm` provides the following results:

```
> summary(lm(y~x))

Call:
lm(formula = y ~ x)

Residuals:
    1      2      3      4      5 
1.48805  0.36860  0.23891 -2.00000 -0.09556 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.52218     2.50349   -3.804  0.03193 *
x              0.48805     0.06041    8.079  0.00396 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.462 on 3 degrees of freedom
Multiple R-squared:  0.9561,    Adjusted R-squared:  0.9414 
F-statistic: 65.26 on 1 and 3 DF,  p-value: 0.003963
```

```
> pMLE
$minimum
[1] 6.338178

$estimate
[1] -9.9213431  0.4977324  1.1377163

$gradient
[1] -1.193328e-09 -3.517187e-08 -1.506689e-09

$hessian
      [,1]      [,2]      [,3]
[1,] 4.45451409 154.511927 0.08146564
[2,] 154.51192681 7744.308141 -5.01151097
[3,] 0.08146564 -5.011511 7.72173440

$code
[1] 1

$iterations
[1] 20

> diag(solve(pMLE$hessian))^0.5
[1] 0.85514563 0.02051154 0.36050402
```

In some sense the prior information improve the inference.

Exercise 2

Remember the example of the blood analysis, an individual is considered healthy if the mean is $H_0: 2 \leq \mu \leq 4$ and infected if $H_1: \mu > 4$.

Assuming that $P(H_0)=P(H_1)=0.5$, the Bayes factor is,

$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{\int_2^4 L(\mu|X) \pi_0(\mu) d\mu}{\int_4^\infty L(\mu|X) \pi_1(\mu) d\mu}$$

Assuming $\pi_0(\mu) = \pi_1(\mu) \sim \text{Gamma}(\alpha, 1)$, where the hyper-parameter α is estimated by the sample mean (4.6), the Bayes factor is,

$$\frac{Pr(H_0|X)}{Pr(H_1|X)} = \frac{\int_2^4 \mu^{463.6} \exp(-101\mu) d\mu}{\int_4^\infty \mu^{463.6} \exp(-101\mu) d\mu}$$

$L(X|\mu)\pi_0(\mu)$ and $L(X|\mu)\pi_1(\mu)$ can be approximated by normal distributions using the Laplace approximation:

$$l^* = 463.6 \log(\mu) - 101\mu ; \quad \frac{dl^*}{d\mu} = \frac{463.6}{\mu} - 101 = 0 \rightarrow \mu^* = 463.6/101 = 4.59$$

$$\frac{d^2 l^*}{d\mu^2} = -\frac{463.6}{\mu^2} \rightarrow I(\mu^*) = 463.6/\mu^{*2} \quad V(\mu^*) = \mu^{*2}/463.6 = 0.04544$$

The normal approximation is $N(4.59, 0.213)$.

It gives a Bayes factor of,

```
> (pnorm(4,4.59,.213)-pnorm(2,4.59,.213))/(1-pnorm(4,4.59,.213))
```

```
[1] 0.002811099
```

It gives an evidence against H_0 of $1/ 0.002811099 = 355.7328$

Note that the exact value was 585.8231.